```
asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ git --version
git version 2.55.0.windows.5

asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ python --version
Python 3.14.6

asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ node -v
npm -v
v24.14.0
11.9.0

asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ go version
go version go1.27.2 windows/amd64

asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ protoc --version
libprotoc 36.2

asus@LAPTOP-4Q1IPJTI MINGW64 ~
$
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ cd ~/Documents

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents
$ mkdir grpc-polyglot-lab

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents
$ cd grpc-polyglot-lab

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab
$ git init
Initialized empty Git repository in C:/Users/asus/Documents/grpc-polyglot-lab/.git/

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab (master)
$ cat > .gitignore <<'EOF'
node_modules/
__pycache__/
.venv/
*.exe
EOF

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab (master)
$ pwd
ls -la
/c/Users/asus/Documents/grpc-polyglot-lab
total 33
drwxr-xr-x 1 asus 197121  0 Oct 15 08:09 ./
drwxr-xr-x 1 asus 197121  0 Oct 15 08:09 ../
drwxr-xr-x 1 asus 197121  0 Oct 15 08:09 .git/
-rw-r--r-- 1 asus 197121 40 Oct 15 08:09 .gitignore
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab (master)
$ cat > user.proto <<'EOF'
syntax = "proto3";

package user;

option go_package = "./pb";

message UserRequest {
  int32 user_id = 1;
}

message UserResponse {
  string name = 1;
  string email = 2;
}

service UserService {
  rpc GetUserProfile(UserRequest) returns (UserResponse);
}
EOF
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab (master)
$ cat user.proto
syntax = "proto3";

package user;

option go_package = "./pb";

message UserRequest {
  int32 user_id = 1;
}

message UserResponse {
  string name = 1;
  string email = 2;
}

service UserService {
  rpc GetUserProfile(UserRequest) returns (UserResponse);
}
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab (master)
$ mkdir go-server
cd go-server
go mod init go-server
go: creating new go.mod: module go-server
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ go get google.golang.org/grpc
go get google.golang.org/protobuf
go: downloading google.golang.org/grpc v1.84.0
go: downloading golang.org/x/net v0.57.0
go: downloading google.golang.org/protobuf v1.36.11
go: downloading golang.org/x/sys v0.47.0
go: downloading google.golang.org/genproto/googleapis/rpc v0.0.0-20260706201446-f0a921348800
go: downloading golang.org/x/text v0.40.0
go: added golang.org/x/net v0.57.0
go: added golang.org/x/sys v0.47.0
go: added golang.org/x/text v0.40.0
go: added google.golang.org/genproto/googleapis/rpc v0.0.0-20260706201446-f0a921348800
go: added google.golang.org/grpc v1.84.0
go: added google.golang.org/protobuf v1.36.11
go: downloading google.golang.org/protobuf v1.36.12
go: upgraded google.golang.org/protobuf v1.36.11 => v1.36.12
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ go install google.golang.org/protobuf/cmd/protoc-gen-go@latest
go install google.golang.org/grpc/cmd/protoc-gen-go-grpc@latest
go: downloading google.golang.org/grpc/cmd/protoc-gen-go-grpc v1.6.2

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ export PATH="$PATH:$(go env GOPATH)/bin"

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ protoc-gen-go --version
protoc-gen-go-grpc --version
protoc-gen-go.exe v1.36.12
protoc-gen-go-grpc 1.6.2
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ mkdir -p pb

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ protoc --proto_path=.. --go_out=pb --go_opt=paths=source_relative --go-grpc_out=pb --go-grpc_opt=paths=source_relative ../user.proto

asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ ls -la pb
total 16
drwxr-xr-x 1 asus 197121    0 Oct 15 08:17 ./
drwxr-xr-x 1 asus 197121    0 Oct 15 08:17 ../
-rw-r--r-- 1 asus 197121 5123 Oct 15 08:17 user.pb.go
-rw-r--r-- 1 asus 197121 4440 Oct 15 08:17 user_grpc.pb.go
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$
$ cat > main.go <<'EOF'
package main

import (
        "context"
        "log"
        "net"

        pb "go-server/pb"

        "google.golang.org/grpc"
        "google.golang.org/grpc/codes"
        "google.golang.org/grpc/status"
)

type server struct {
        pb.UnimplementedUserServiceServer
}

func (s *server) GetUserProfile(ctx context.Context, req *pb.UserRequest) (*pb.UserResponse, error) {
        log.Printf("Request diterima untuk user_id=%d", req.GetUserId())

        if req.GetUserId() == 1 {
                return &pb.UserResponse{
                        Name:  "Budi Santoso",
                        Email: "budi@universitas.ac.id",
                }, nil
        }

        return nil, status.Error(codes.NotFound, "user tidak ditemukan")
}

func main() {
        listener, err := net.Listen("tcp", ":50051")
        if err != nil {
                log.Fatalf("Gagal membuka port: %v", err)
        }

        grpcServer := grpc.NewServer()
        pb.RegisterUserServiceServer(grpcServer, &server{})

        log.Println("gRPC server berjalan di port 50051")

        if err := grpcServer.Serve(listener); err != nil {
                log.Fatalf("Server gagal: %v", err)
        }
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (master)
$ go mod tidy
go run main.go
go: downloading github.com/google/go-cmp v0.7.0
go: downloading github.com/golang/protobuf v1.5.4
2026/10/15 08:19:15 gRPC server berjalan di port 50051
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~
$ cd ~/Documents/grpc-polyglot-lab/python-client
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/python-client (master)
$ source .venv/Scripts/activate
(.venv)
```

```
(.venv)
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/python-client (master)
$ python -m pip show grpcio grpcio-tools
Name: grpcio
Version: 1.84.0
Summary: HTTP/2-based RPC framework
Home-page: https://grpc.io
Author:
Author-email: The gRPC Authors <grpc-io@googlegroups.com>
License-Expression: Apache-2.0
Location: C:\Users\asus\Documents\grpc-polyglot-lab\python-client\.venv\Lib\site-packages
Requires: typing-extensions
Required-by: grpcio-tools
---
Name: grpcio-tools
Version: 1.84.0
Summary: Protobuf code generator for gRPC
Home-page: https://grpc.io
Author:
Author-email: The gRPC Authors <grpc-io@googlegroups.com>
License-Expression: Apache-2.0
Location: C:\Users\asus\Documents\grpc-polyglot-lab\python-client\.venv\Lib\site-packages
Requires: grpcio, protobuf, setuptools
Required-by:
(.venv)
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/python-client (master)
$ python -m pip install grpcio grpcio-tools
Requirement already satisfied: grpcio in .\.venv\Lib\site-packages (1.84.0)
Requirement already satisfied: grpcio-tools in .\.venv\Lib\site-packages (1.84.0)
Requirement already satisfied: typing-extensions~=4.12 in .\.venv\Lib\site-packages (from grpcio) (4.16.0)
Requirement already satisfied: protobuf<8.0.0,>=7.35.1 in .\.venv\Lib\site-packages (from grpcio-tools) (7.36.2)
Requirement already satisfied: setuptools>=77.0.1 in .\.venv\Lib\site-packages (from grpcio-tools) (84.0.0)
(.venv)
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/python-client (master)
```

client.py | main.go | client.js

python-client > client.py > ...

```python
import grpc
import user_pb2
import user_pb2_grpc


def main():
    with grpc.insecure_channel("localhost:50051") as channel:
        stub = user_pb2_grpc.UserServiceStub(channel)

        request = user_pb2.UserRequest(user_id=1)
        response = stub.GetUserProfile(request)

        print("Nama:", response.name)
        print("Email:", response.email)


if __name__ == "__main__":
    main()
```

```
(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab\python-client> python client.py
Nama: Budi Santoso
Email: budi@universitas.ac.id
```

```
(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab> mkdir node-client


    Directory: C:\Users\asus\Documents\grpc-polyglot-lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/15/2026   9:57 AM                node-client


(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab> cd node-client
```

```
(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab\node-client> npm init -y
Wrote to C:\Users\asus\Documents\grpc-polyglot-lab\node-client\package.json:

{
  "name": "client",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}


(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab\node-client> npm install @grpc/grpc-js @grpc/proto-loader

added 33 packages, and audited 34 packages in 13s

3 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

client.py | main.go | client.js

node-client > client.js > ...

```js
const grpc = require("@grpc/grpc-js");
const protoLoader = require("@grpc/proto-loader");

const PROTO_PATH = "../user.proto";

const packageDefinition = protoLoader.loadSync(PROTO_PATH, {
    keepCase: true,
    longs: String,
    enums: String,
    defaults: true,
    oneofs: true,
});

const protoDescriptor = grpc.loadPackageDefinition(packageDefinition);
const userPackage = protoDescriptor.user;

const client = new userPackage.UserService(
    "localhost:50051",
    grpc.credentials.createInsecure()
);

client.GetUserProfile({ user_id: 1 }, (error, response) => {
    if (error) {
        console.error("Error:", error.message);
        client.close();
        return;
    }

    console.log("Nama:", response.name);
    console.log("Email:", response.email);

    client.close();
});
```

```
(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab\node-client> node client.js
Nama: Budi Santoso
Email: budi@universitas.ac.id
(.venv) PS C:\Users\asus\Documents\grpc-polyglot-lab\node-client>
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/go-server (main)
$ go run main.go
2026/10/19 00:49:02 gRPC server berjalan di port 50051
2026/10/19 00:57:16 Request diterima untuk user_id=1
2026/10/19 01:00:30 Request diterima untuk user_id=1
exit status 0xc000013a
```

```
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/python-client (main)
$ python client.py
Nama: Nisa Tridasari
Email: nisa@universitas.ac.id
(.venv)
asus@LAPTOP-4Q1IPJTI MINGW64 ~/Documents/grpc-polyglot-lab/python-client (main)
$
```

